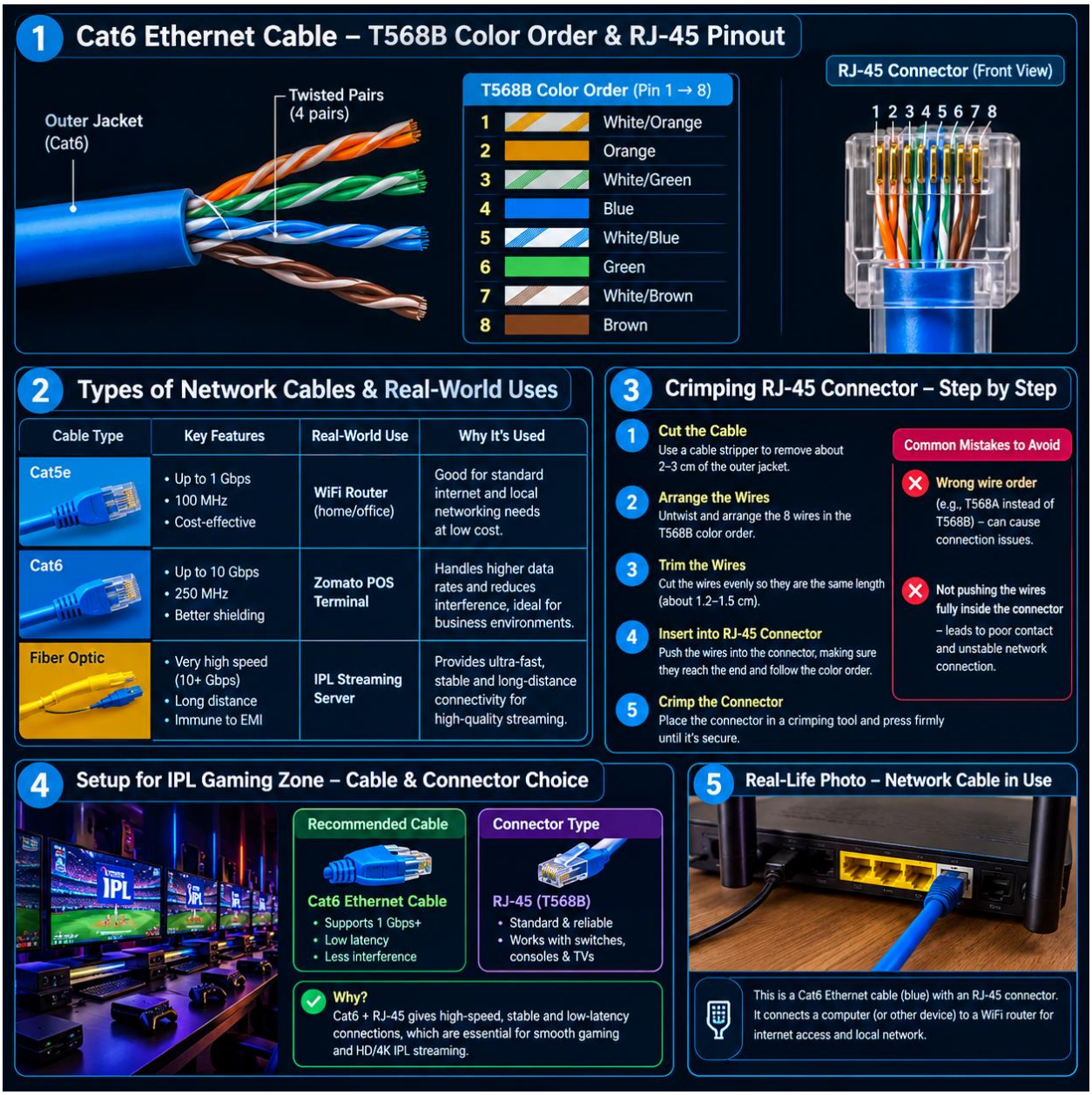
1 Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



1. Cat6 Ethernet Cable – T568B Standard

The T568B standard defines the wire color order inside an RJ-45 connector. It contains 8 pins and 4 twisted pairs.

Pin No.	Wire Color
1	White-Orange
2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

2 List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable Type	Real-World Application	Reason
Cat5e	Wi-Fi Router to Computer	Cost-effective and supports speeds up to 1 Gbps.
Cat6	Zomato POS Terminal	Provides high-speed data transmission with reduced interference.
Fiber Optic	IPL Streaming Server	Supports very high speeds and long-distance communication.

3 Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Steps to Crimp an RJ-45 Connector

1. Cut the cable: Cut the Ethernet cable to the required length.
2. Strip the outer jacket: Remove about 2–3 cm of the outer covering using a cable stripper.
3. Arrange the wires: Arrange all eight wires according to the T568B color standard.
4. Trim the wires: Cut the wires evenly so they fit inside the RJ-45 connector.
5. Insert the wires: Push all eight wires fully into the connector in the correct order.
6. Crimp the connector: Use a crimping tool to secure the connector onto the cable.
7. Test the cable: Use a LAN cable tester to verify that all eight pins connect correctly.

Two common mistakes to avoid:

- Arranging the wires in the wrong color order.
- Not inserting the wires fully into the RJ-45 connector before crimping.

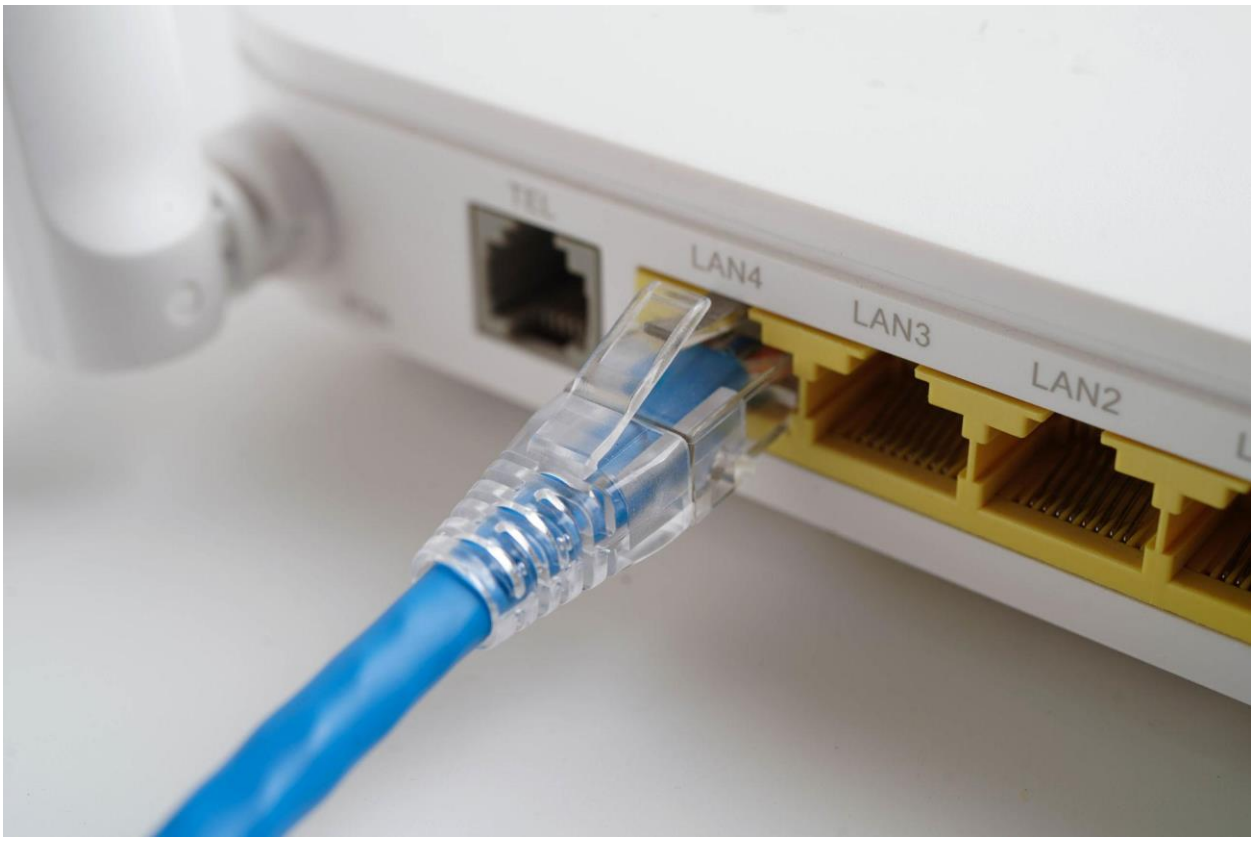
4 Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cable Selection for an IPL Gaming Zone

- Cable: Cat6 Ethernet cable.
- Connector: RJ-45 connector using the T568B standard.
- Reason: Cat6 supports high-speed connections up to 10 Gbps over suitable distances, with low latency and reduced interference. It is suitable for connecting gaming consoles, computers, and network switches for reliable gaming and streaming.
- For long distances: Fiber optic cable can connect network switches between distant areas of the event venue.

5 Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Description:

- Cable type: Cat6 Ethernet cable.
- Connector: RJ-45 connector.
- Connected devices: Wi-Fi router and computer or network switch.
- Purpose: It provides a wired network connection for reliable internet access and data transfer.